

Advanced Networking

Topic 1: TCP vs UDP

TCP (Transmission Control Protocol) is connection-oriented, providing reliable, ordered, and error-checked delivery of a stream of octets. UDP (User Datagram Protocol) is connectionless, offering faster but unreliable delivery, often used for live streaming or gaming.

Topic 2: Ports

A port is a logical construct that identifies a specific process or a type of network service. Port numbers range from 0 to 65535. For example, HTTP uses port 80, HTTPS uses port 443, and SSH uses port 22.

Topic 3: Subnetting

Subnetting is the practice of dividing a network into two or more smaller networks. It improves network performance and security. A subnet mask is used to distinguish the network portion of an IP address from the host portion.